

1 Apparent Catamenial Epilepsy Classification Under a Null Simulation

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Abstract

Objective: To quantify how often catamenial epilepsy classification rules produce apparent catamenial epilepsy under a null model in which seizure diaries and menstrual-cycle diaries are generated independently. Methods: We simulated 100,000 synthetic participants for 36 months in healthy ovulatory and heterogeneous menstruating-age cohorts. CHOCOLATES generated seizure diaries and HORMONE-CYCLE generated independent menstrual/hormone diaries. A random circular shift was applied before merging. Herzog phases were labeled before window sampling. We evaluated exact Herzog 2004, windowed Herzog thresholds, minimum-data rules, cycle reproducibility rules, stabilized-dispersion and window-dispersion negative-binomial regression, and historical-rule operationalizations across defined calendar, cycle, and full-diary windows. Results: In 36-month windows, windowed Herzog thresholds classified 11.2% (Monte Carlo binomial 95% interval 11.0-11.5) of healthy ovulatory and 36.3% (35.8-36.7) of heterogeneous cohort participants as apparent catamenial epilepsy under independence. Stabilized-dispersion negative-binomial regression exhibited approximately nominal Type I error, 4.1% (3.9-4.2) and 4.2% (4.0-4.4), respectively. Exact Herzog 2004 applied to 3 complete cycles classified 50.3% and 51.6% of classifiable windows positive, with 53.8% and 66.7% not classifiable. In 10,000 null studies of 30 participants using 3-month windows, apparent prevalence exceeded 39.1% in 45.9% of healthy ovulatory and 79.9% of heterogeneous cohort studies. Conclusions: These findings quantify the potential for false-positive catamenial epilepsy classification under finite diary windows and independence between seizure and menstrual processes. Diagnostic studies should prespecify analyzability, comparator handling, ovulatory-status handling, and reproducibility requirements.

47 Classification of Evidence: Not applicable. This simulation study evaluates operating
48 characteristics of classification rules under a synthetic null model and does not test a diagnostic
49 intervention or therapeutic effect in human participants.

50 Search Terms: catamenial epilepsy; epilepsy; seizures; menstrual cycle; simulation study.

51

52 **Introduction**

53 Catamenial epilepsy describes seizure exacerbation in relation to the menstrual cycle. The
54 clinical literature has long recognized perimenstrual, periovulatory, and inadequate luteal-phase
55 patterns, but reported prevalence varies widely because studies use different phase definitions,
56 thresholds, diary durations, and eligibility rules. Herzog and colleagues reported 39.1%
57 catamenial epilepsy in women who charted seizures and menses for three complete cycles using
58 a two-of-three-cycle rule, and later work from the NIH Progesterone Treatment Trial has been
59 interpreted as supporting prevalence near 44% in selected trial participants.^{1,5}

60 The diagnostic problem is not only biological. Seizures are temporally structured: clustering,
61 circadian rhythms, and multidien rhythms have been documented in long-term diaries and
62 intracranial recordings.⁹⁻¹¹ If seizure rhythms are independent of menstrual cycles but contain
63 multidien structure in overlapping period bands, finite diary windows can still contain chance
64 alignments. Ratio definitions are especially vulnerable when seizure counts are sparse or
65 comparator phases contain few events. This creates a specificity question with direct
66 implications for clinical research: what apparent catamenial epilepsy rate should be expected
67 when no menstrual coupling exists?

68 We conducted a reproducible null-simulation study to estimate apparent catamenial epilepsy
69 classification across definitions and observation windows under independence. The analysis
70 evaluates specificity-like behavior only. Positive-control true-coupling simulations were added
71 as a sensitivity framework to calibrate future sensitivity analyses, but the primary inference is the
72 null background rate.

73 **Methods**

74 Study design. This Monte Carlo null simulation was defined before the full simulation run. For
75 each synthetic participant, seizure and menstrual/hormone diaries were generated independently
76 for 36 months. A participant-specific random circular shift of the seizure diary was sampled
77 uniformly over the diary length before merging to prevent hidden start-date alignment; cycle
78 labels were preserved from the hormone diary and seizure counts were wrapped with the shifted
79 seizure diary. All analyses were stratified by cohort; cohorts were never pooled.

80 Cohorts. The healthy ovulatory cohort included 50,000 synthetic participants, age 18-45 years,
81 with ovulatory cycling enforced through the simulator adapter where available. The
82 heterogeneous menstruating-age cohort included 50,000 synthetic participants, age 13.0-54.9
83 years, allowing anovulation and irregularity. Because the hormone-cycle simulator exposed
84 medical-factor controls but not a natural prevalence sampler, PCOS, peri-menarche,
85 perimenopause, and dysmenorrhea were sampled from configuration rates. This cohort should be
86 interpreted as an assumption-driven heterogeneity stress test unless these rates are replaced with
87 externally validated prevalence inputs.

88 Diary simulators. CHOCOLATES generated non-catamenial seizure diaries using its default
89 seizure-burden, clustering, and multidien-cycle behavior. HORMONE-CYCLE generated
90 independent menstrual/hormone diaries, including cycle length, ovulatory status, luteal
91 progesterone, and inadequate-luteal-phase flags where exposed. Adapter functions recorded
92 simulator outputs and latent summaries, including seizure burden, dominant seizure-cycle period
93 when available, cycle-length summaries, ovulatory fraction, and medical-factor flags. Simulator
94 validation targets, versions, and package commits are documented in the Appendix following
95 STRESS reporting principles.¹²

96 Phase labeling and definitions. Herzog phases were assigned on the full diary before subsetting.
97 In a complete cycle of length L , day 1 was menstrual-flow onset and backward day was $d-(L+1)$.
98 Labels were assigned in the following order: menstrual phase if forward day 1-3 or backward
99 day -3 to -1; follicular phase if forward day 4-9; ovulatory phase if forward day ≥ 10 and
100 backward day ≤ -13 ; luteal phase if backward day -12 to -4; otherwise unlabeled. The ovulatory
101 condition is an intersection, not a union. Windowed thresholds used $C1 = \text{ADSF}(M)/\text{ADSF}(F+L)$
102 ≥ 1.69 , $C2 = \text{ADSF}(O)/\text{ADSF}(F+L) \geq 1.83$, and $C3 = \text{ADSF}(O+L+M)/\text{ADSF}(F) \geq 1.62$
103 among inadequate-luteal-phase days. $C3$ was evaluated only in the heterogeneous cohort when
104 `ilp_flag` was present; anovulatory or nonclassifiable cycles without `ilp_flag` did not contribute to
105 $C3$. $C3$ -excluding and $C1/C2/C3$ -only sensitivity endpoints were generated.
106 Exact Herzog 2004 was restricted to complete 3-cycle windows with all cycles 23-35 days and
107 required at least two of three cycles to show any positive pattern. Other calendar and longer
108 cycle windows are reported as not evaluated for this rule rather than interpreted as negative.
109 Minimum-data rules required at least 4 calendar months or 6 complete cycles and at least 4
110 seizure days. Reproducibility rules required at least two-thirds of eligible cycles to be positive
111 for the same pattern and the corresponding pooled ratio to pass threshold.
112 Regression comparator. The primary regression comparator used a within-participant negative-
113 binomial generalized linear model with daily seizure count as outcome, log link, intercept,
114 menstrual-phase indicator, ovulatory-phase indicator, and cycle fixed effects when at least four
115 complete cycles were available. The offset was one observed day for each daily row. One-sided
116 Wald tests evaluated menstrual and ovulatory enrichment, Holm-adjusted across the two
117 comparisons; positivity required both adjusted $P < 0.05$ and the Herzog rate-ratio threshold for
118 that phase. $C3$ was not evaluated by regression. The primary regression used full-diary method-

of-moments dispersion as a stabilized-dispersion comparator, and a window-only dispersion sensitivity was added. If model fitting failed, a robust Poisson fallback was recorded. Statistical analysis. The primary outcome was the person-window apparent catamenial epilepsy rate among classifiable windows under the null. We also report positivity among all attempted windows when indeterminate rates are material. Monte Carlo binomial 95% intervals quantify simulation uncertainty under the fixed model and do not incorporate uncertainty about simulator form, medical-factor rates, or period distributions. Study-level Monte Carlo independently sampled 10,000 studies of 30 participants without replacement from each cohort, selecting one precomputed random valid 3-month window per participant, and estimated probabilities that apparent prevalence exceeded 39.1% and 44.2%. Positive-control sensitivity. A configurable true-coupling mode adds extra Poisson seizure events after null diary alignment on target phase days. The current positive-control setting targets menstrual-phase days with a 1.5-fold rate-ratio input. This is an auditable operating-characteristic sensitivity rather than a mechanistic hormone model. Standard protocol approvals, registrations, and patient consents. This study used only synthetic participants and did not use human-subject data, identifiable patient information, images, or videos. Institutional review board approval and patient consent were not required.

Results

Simulation cohort and diary characteristics. The full run included 100,000 synthetic participants and 900,000 defined participant-window rows. Cohort summaries are shown in Table 1. The healthy ovulatory cohort had 100.0% ovulatory cycles by design. The heterogeneous cohort had 79.0% ovulatory cycles, greater cycle-length variability, and similar observed seizure burden.

141 Primary full-diary null rates. In 36-month windows, the windowed Herzog threshold definition
142 classified 11.2% of healthy ovulatory participants and 36.3% of heterogeneous cohort
143 participants as apparent catamenial epilepsy under independence (Table 2). Adding minimum-
144 data rules changed the full-window estimates modestly because most full diaries were
145 analyzable. Stabilized-dispersion negative-binomial regression exhibited approximately nominal
146 Type I error at 4.1% and 4.2%, as expected for a calibrated test under the null. This calibration
147 should not be interpreted as clinical usefulness; sensitivity and positive predictive value require
148 true-coupling simulations.

149 C3 contribution. In the heterogeneous cohort, the high windowed full-diary null rate was driven
150 largely by C3 logic. Full-window C3 positivity occurred in 37.0% of applicable heterogeneous-
151 cohort windows, compared with C1 and C2 rates of 8.4% and 5.6%. C3-excluding endpoints and
152 C1/C2/C3-only sensitivity rows are reported in the Appendix to separate inadequate-luteal-phase
153 logic from perimenstrual and periovulatory classification.

154 Observation-window effects. Short diary windows were the most vulnerable to apparent
155 catamenial classification. In random 3-month calendar windows, windowed Herzog thresholds
156 yielded apparent classification rates of 41.3% in the healthy ovulatory cohort and 51.2% in the
157 heterogeneous cohort. Minimum-data, reproducibility, and regression definitions were
158 intentionally not classifiable for 3-month calendar windows when the required minimum
159 structure was absent. Exact Herzog 2004 applied to 3 complete cycles yielded apparent
160 classification rates of 50.3% and 51.6% among classifiable windows, while 53.8% and 66.7% of
161 attempted 3-cycle windows were not classifiable.

162 Study-level Monte Carlo. In 10,000 simulated studies of 30 participants using 3-month windows
163 and the windowed Herzog threshold rule, the mean apparent prevalence among all 30

participants was 37.5% in the healthy ovulatory cohort and 46.0% in the heterogeneous cohort.

Apparent prevalence was at least 39.1% in 45.9% and 79.9% of null studies, respectively, and at least 44.2% in 19.7% and 54.5%.

Daily audit profile. The cycle-day audit plot is retained as an exploratory quality-control display rather than primary evidence. Because late cycle days have sparse denominators in irregular cycles, interpretation should emphasize phase-level and window-level endpoints rather than isolated day-level spikes.

Discussion

This null simulation shows that apparent catamenial epilepsy can arise frequently even when seizure and menstrual processes are generated independently. The magnitude depends strongly on the definition, window length, analyzability rules, and cohort menstrual-cycle assumptions.

Ratio-only windowed definitions were vulnerable, particularly in short windows and in the heterogeneous cohort where inadequate-luteal-phase logic made C3 classification common.

Regression and reproducibility requirements reduced null positives but did so partly by requiring longer diaries or by declining to classify insufficient windows.

The negative-binomial result should be interpreted as calibration, not superiority. A statistical test evaluated at $\alpha=0.05$ should produce approximately 5% positives under a correctly specified null, and the stabilized-dispersion regression did so. Ratio rules are not nominal-alpha tests and therefore have no comparable Type I error guarantee. Whether regression, reproducibility rules, or ratio thresholds are useful for diagnosis depends on sensitivity, positive predictive value, burden, interpretability, and robustness under true coupling. The positive-control framework added here is intended to support that operating-characteristic assessment.

186 These results help interpret the broad range of prevalence estimates in the catamenial epilepsy
187 literature without estimating true prevalence. Earlier work established biologically plausible and
188 clinically important menstrual-cycle seizure patterns, including C1, C2, and C3 patterns.¹⁻⁵
189 Treatment literature also suggests that robust perimenstrual exacerbation may identify subgroups
190 of interest, even though randomized evidence for hormonal therapy remains limited and
191 mixed.^{6,7} The present analysis instead quantifies a diagnostic background rate expected from
192 chance alignment when seizure cycles and menstrual cycles coexist under independence.
193 The findings are consistent with modern seizure-cycle research. Multidien seizure rhythms and
194 seizure forecasting studies have shown that seizure risk varies over days to weeks, and those
195 rhythms can be detected in long-term electronic diaries and chronic recordings.⁹⁻¹¹ A
196 menstrual-cycle analysis that does not account for intrinsic seizure cyclicity can therefore
197 mistake phase overlap for hormone coupling. This is most problematic when the observation
198 window is short, because a few events can dominate average daily seizure-frequency ratios.
199 Several methodologic implications follow. First, catamenial epilepsy studies should report both
200 positive rates and indeterminate rates, because stricter definitions may appear more conservative
201 partly by declining to classify insufficient windows. Second, comparator phases with zero
202 seizures require explicit rules; silently treating undefined ratios as negative can bias estimates.
203 Third, benchmark comparisons should match the original study design. For Herzog 2004, the
204 design-matched comparison is the exact 3-complete-cycle rule, whereas random 3-calendar-
205 month windows are a design-sensitivity analysis. Fourth, C3 classification in heterogeneous or
206 irregular-cycle populations requires careful ovulatory-status and luteal-phase ascertainment.
207 This study has limitations. The results are only as credible as the CHOCOLATES and
208 HORMONE-CYCLE simulators, their parameter settings, and the adapter assumptions. The

heterogeneous cohort medical-factor rates were configuration-sampled rather than drawn from a simulator-native prevalence sampler, and cycle-length variability is therefore a sensitivity driver rather than a settled population estimate. Monte Carlo binomial intervals do not include model-form uncertainty. The current positive-control coupling model is simple and perimenstrual; it does not represent full hormone dynamics, treatment effects, wavelet analyses, hidden Markov models, medication changes, or adjudicated seizure types.

In conclusion, under an independence model with structured seizure diaries and menstrual-cycle diaries, apparent catamenial epilepsy classification can be common and can reach historical benchmark prevalence values. Future diagnostic and interventional studies should prespecify diary length, menstrual-phase labeling, analyzability thresholds, zero-denominator handling, ovulatory-status handling, and reproducibility criteria before interpreting apparent menstrual clustering as evidence of hormone-linked seizure exacerbation.

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Disclosure

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230 Data Availability

231 The simulation code, configuration, analysis outputs, and machine-readable manifest are
232 available in the project repository. The reproducible run used master seed 20260505 and the
233 manifest in outputs/manifest.json records config parameters, output paths, file sizes, and
234 SHA-256 checksums. A versioned public archive with DOI will accompany the submitted
235 version.

236

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270 Tables

271 Table 1. Simulated cohort characteristics.

Cohort	N	Mean age, y	Mean cycle length, d	Cycle-length SD, d	Ovulatory cycles	Seizure days/month	Seizures/month
healthy ovulatory	50,000	31.5 (SD 7.8)	29.15	3.39	100.0%	2.46	6.83
heterogeneous menstruating-age	50,000	34.0 (SD 12.1)	30.92	4.74	79.0%	2.45	6.80

272

273 Values are cohort-level means from the full 100,000-participant run. Cycle-length SD is the

274 mean within-participant cycle-length SD.

275 Table 2. Key null false-positive results.

Cohort	Analysis setting	Definition	Classifiable denominator	Main result	Indeterminate	Benchmark probabilities
healthy ovulatory	36-month full diary	Windowed Herzog thresholds	49,605 / 50,000	11.2% (11.0%-11.5%)	0.8%	Not applicable
healthy ovulatory	36-month full diary	Windowed Herzog with minimum data	48,359 / 50,000	10.2% (9.9%-10.5%)	3.3%	Not applicable
healthy ovulatory	36-month full diary	Negative-binomial regression, stabilized dispersion	48,359 / 50,000	4.1% (3.9%-4.2%)	3.3%	Not applicable
healthy ovulatory	3 complete cycles	Exact Herzog 2004, any pattern	23,101 / 50,000	50.3% (49.6%-50.9%)	53.8%	Not applicable
healthy ovulatory	10,000 n=30 studies; 3-month windows	Windowed Herzog thresholds	All 30 participants	Mean apparent prevalence 37.5%	Not applicable	45.9%; 19.7%
heterogeneous menstruating-age	36-month full diary	Windowed Herzog thresholds	49,587 / 50,000	36.3% (35.8%-36.7%)	0.8%	Not applicable
heterogeneous menstruating-age	36-month full diary	Windowed Herzog with minimum data	48,333 / 50,000	35.6% (35.2%-36.0%)	3.3%	Not applicable

Cohort	Analysis setting	Definition	Classifiable denominator	Main result	Indeterminate	Benchmark probabilities
heterogeneous menstruating-age	36-month full diary	Negative-binomial regression, stabilized dispersion	48,333 / 50,000	4.2% (4.0%-4.4%)	3.3%	Not applicable
heterogeneous menstruating-age	3 complete cycles	Exact Herzog 2004, any pattern	16,673 / 50,000	51.6% (50.8%-52.4%)	66.7%	Not applicable
heterogeneous menstruating-age	10,000 n=30 studies; 3-month windows	Windowed Herzog thresholds	All 30 participants	Mean apparent prevalence 46.0%	Not applicable	79.9%; 54.5%

276

277 Main result is false-positive rate for participant windows unless the row is labeled as a study-

278 level Monte Carlo result. Benchmark probabilities are P(apparent prevalence at least 39.1%) and

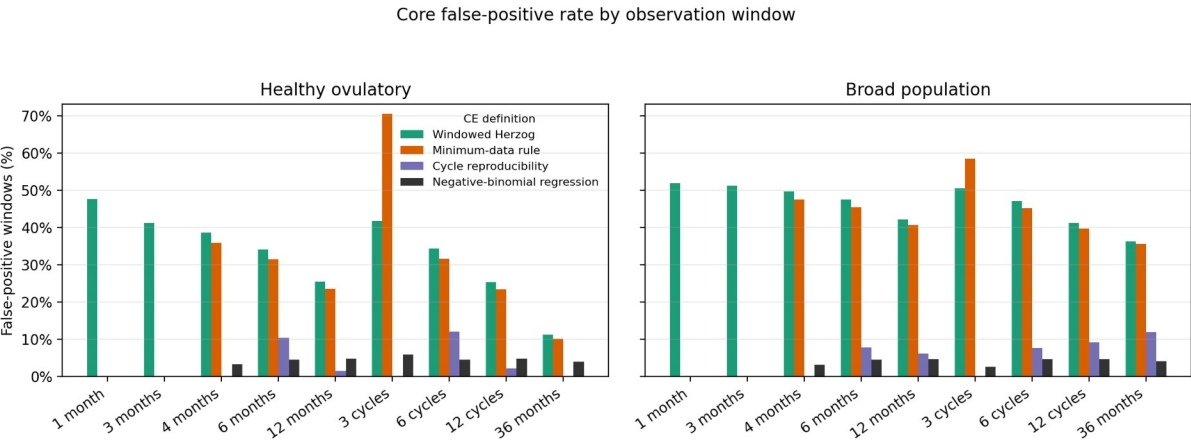
279 P(apparent prevalence at least 44.2%).

280

281 **Figure Legends**

282 Figure 1. False-positive rates by observation window.

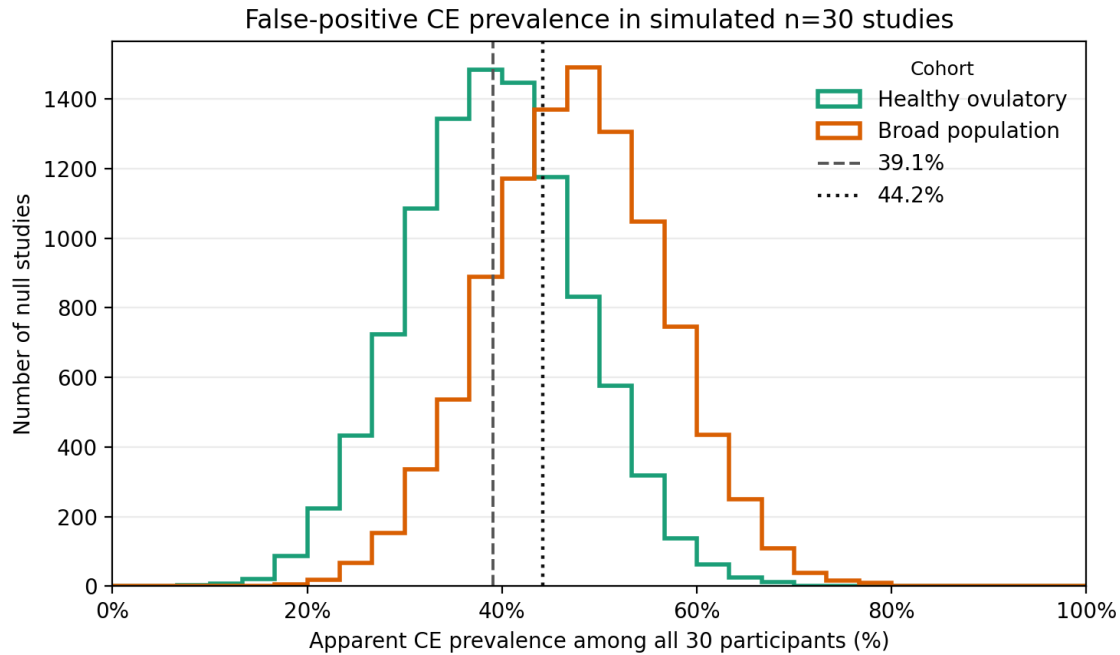
283 Bars show the percentage of classifiable participant windows classified as catamenial epilepsy
284 under the null. Panels are separated by cohort; definitions are shown in the legend.



285

286 Figure 2. Apparent prevalence in null studies.

287 Distribution of apparent catamenial epilepsy prevalence in 10,000 simulated studies of 30
288 participants using random 3-month windows and the windowed Herzog threshold definition.
289 Vertical lines show benchmark prevalence values of 39.1% and 44.2%.

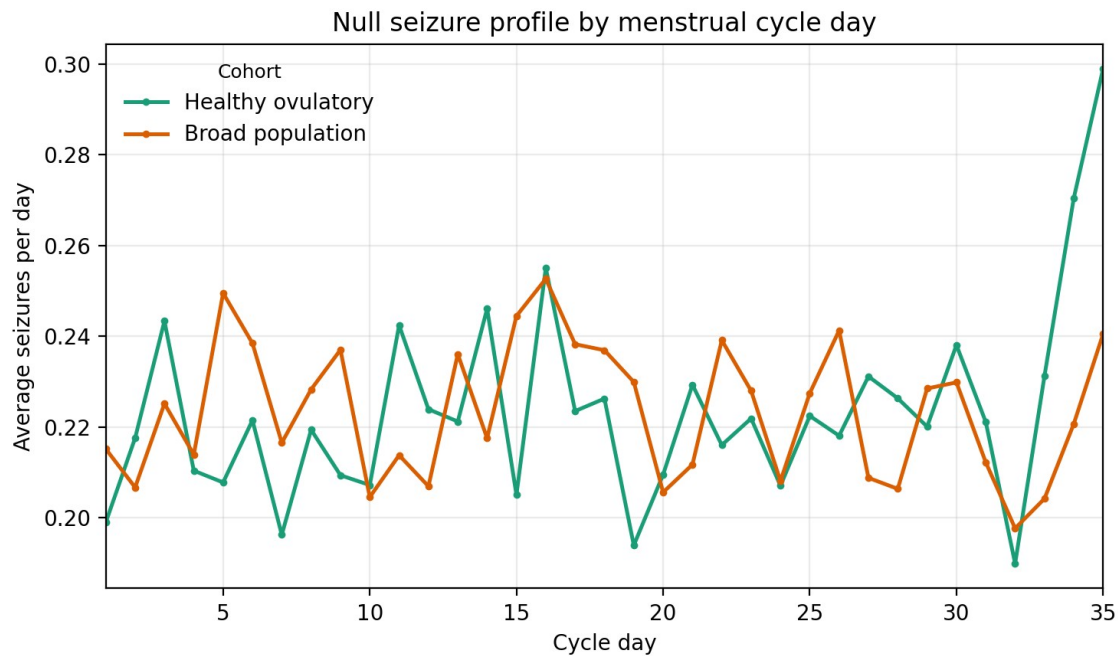


290

291 Figure 3. Null seizure profile by cycle day.

292 Average daily seizure frequency by menstrual cycle day in the 1% daily audit sample. The

293 independent null simulation should not produce a coherent hormone-coupled seizure profile.



294